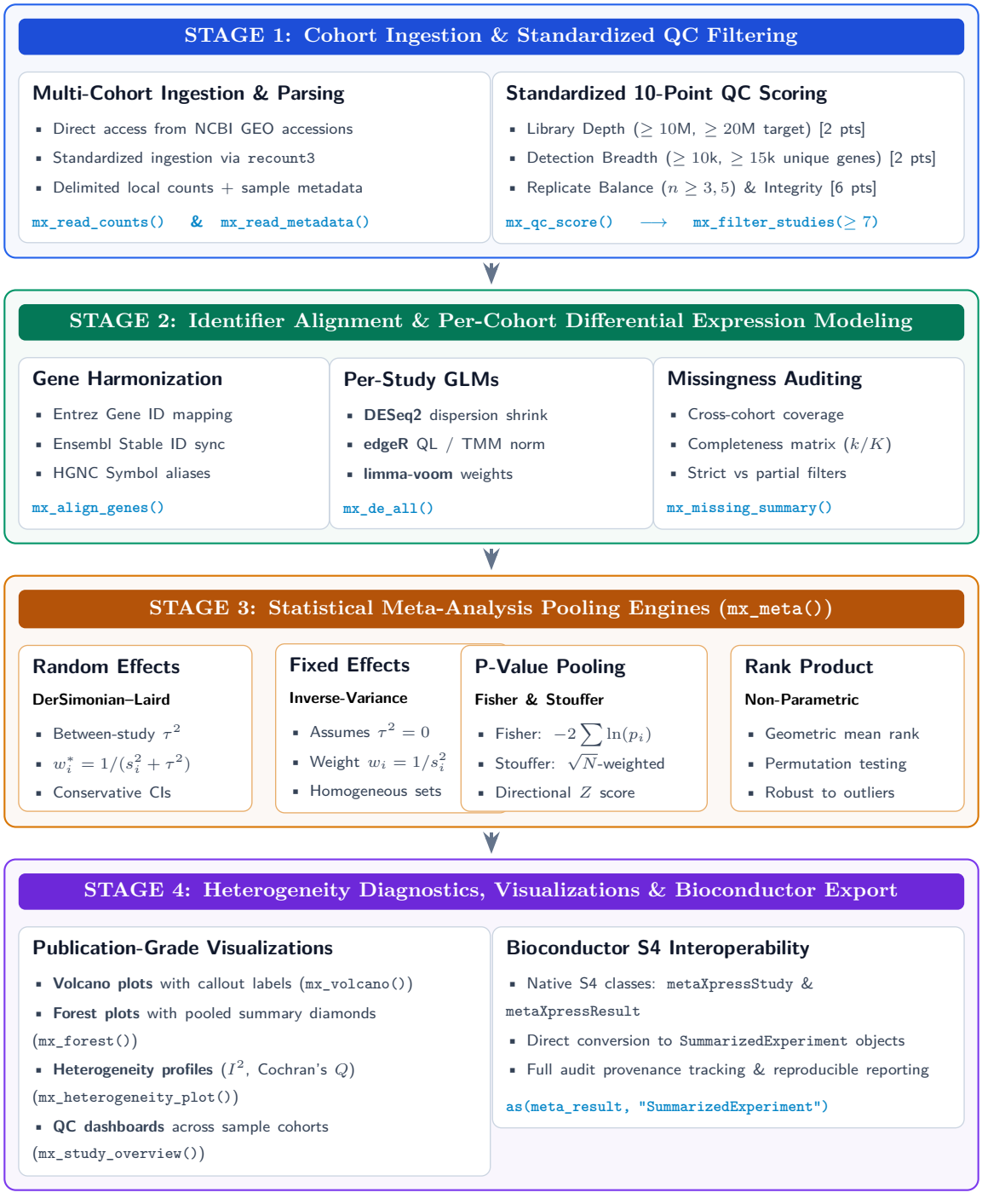

STAGE 1: Cohort Ingestion & Standardized QC Filtering
Multi-Cohort Ingestion & Parsing
Direct access from NCBI GEO accessions
Standardized ingestion via recount3
Delimited local counts + sample metadata
mx_read_counts() & mx_read_metadata()
Standardized 10-Point QC Scoring
Library Depth (≥ 10M, ≥ 20M target) [2 pts]
Detection Breadth (≥ 10k, ≥ 15k unique genes) [2 pts]
Replicate Balance (n ≥ 3, 5) & Integrity [6 pts]
mx_qc_score() ⟶ mx_filter_studies(≥ 7)
STAGE 2: Identifier Alignment & Per-Cohort Differential Expression Modeling
Gene Harmonization
Entrez Gene ID mapping
Ensembl Stable ID sync
HGNC Symbol aliases
mx_align_genes()
Per-Study GLMs
DESeq2 dispersion shrink
edgeR QL / TMM norm
limma-voom weights
mx_de_all()
Missingness Auditing
Cross-cohort coverage
Completeness matrix (k/K)
Strict vs partial filters
mx_missing_summary()
STAGE 3: Statistical Meta-Analysis Pooling Engines (mx_meta())
Random Effects
DerSimonian–Laird
Between-study τ²
w*ᵢ = 1/(s²ᵢ + τ²)
Conservative CIs
Fixed Effects
Inverse-Variance
Assumes τ² = 0
Weight wᵢ = 1/s²ᵢ
Homogeneous sets
P-Value Pooling
Fisher & Stouffer
Fisher: −2 ∑ ln(pᵢ)
Stouffer: √N-weighted
Directional Z score
Rank Product
Non-Parametric
Geometric mean rank
Permutation testing
Robust to outliers
STAGE 4: Heterogeneity Diagnostics, Visualizations & Bioconductor Export
Publication-Grade Visualizations
Volcano plots with callout labels (mx_volcano())
Forest plots with pooled summary diamonds (mx_forest())
Heterogeneity profiles (I², Cochran's Q) (mx_heterogeneity_plot())
QC dashboards across sample cohorts (mx_study_overview())
Bioconductor S4 Interoperability
Native S4 classes: metaXpressStudy & metaXpressResult
Direct conversion to SummarizedExperiment objects
Full audit provenance tracking & reproducible reporting
as(meta_result, "SummarizedExperiment")